Day 2: Test Scenarios and Test Case Design

Requirements breakdown, test case anatomy, Boundary Value Analysis, Equivalence Partitioning, and RTM.

1. Day Learning Objectives

[ ] Transform business requirements into detailed, unambiguous test scenarios
[ ] Structure professional test cases with standard fields (ID, preconditions, test data, steps, expected vs actual)
[ ] Apply Equivalence Partitioning (EP) to mathematically reduce redundant test combinations
[ ] Apply Boundary Value Analysis (BVA) using 2-point and 3-point boundary techniques
[ ] Construct a Decision Table for complex multi-condition business logic
[ ] Maintain a Requirements Traceability Matrix (RTM) to ensure 100% test coverage

2. Core Concepts & Practical Fundamentals

1. Requirements Analysis & Test Scenarios vs Test Cases

Requirements Analysis is the process of reviewing user stories, acceptance criteria, and wireframes to discover ambiguities, missing edge conditions, and technical constraints.

• Test Scenario: A high-level description of what to test (one-line scope, e.g., "Verify user can log in with valid credentials").
• Test Case: A low-level, step-by-step procedure detailing how to test, including specific test data, preconditions, explicit steps, and exact expected outcomes.

Real-World Workplace Examples:
• **[Ride-Hailing Application]:** Test Scenario: Verify passenger can apply promo coupon.
Test Case: Navigate to payment screen, enter code "SAVE20", verify 20% deduction appears, confirm order total decreases from $50.00 to $40.00.

Key Standard Terminology:
• **Acceptance Criteria (AC):** Conditions that a software product must satisfy to be accepted by a user or product owner.
• **Test Scenario:** A generalized test condition representing a functional flow or user journey.
• **Test Case:** A formal artifact specifying inputs, execution conditions, testing procedure, and expected results.

Common Beginner Mistakes & Interview Traps:
& Writing vague expected results such as "It should work properly" instead of explicit criteria like "Displays green banner 'Password updated'".
& Omitting necessary preconditions (e.g. failing to mention user must have $0 balance before testing low-balance alert).

2. Equivalence Partitioning (EP) & Boundary Value Analysis (BVA)

• Equivalence Partitioning (EP): Divides input data into valid and invalid partitions. Testing one representative value from each partition gives the same defect detection confidence as testing every value in that partition.

• Boundary Value Analysis (BVA): Software faults cluster heavily around boundaries of input ranges (due to developer off-by-one errors like using > instead of >=). Test the exact boundary edges:
- Minimum value (Min)
- Just below minimum (Min - 1)
- Just above minimum (Min + 1)
- Maximum value (Max)
- Just below maximum (Max - 1)
- Just above maximum (Max + 1)

Real-World Workplace Examples:
• **[Car Rental Age Eligibility: 21 to 70 years]:** Valid Partition: 21 to 70 (e.g., 35)
Invalid Partition 1: < 21 (e.g., 15)
Invalid Partition 2: > 70 (e.g., 85)
Boundaries: 20 (invalid), 21 (valid), 22 (valid), 69 (valid), 70 (valid), 71 (invalid).

Key Standard Terminology:
• **Equivalence Partition:** A portion of an input domain where all values are expected to be processed similarly by the software.
• **Boundary Value:** A minimum or maximum value at the edge of an equivalence partition.

Common Beginner Mistakes & Interview Traps:
& Testing only middle values (e.g., age 30 and 45) and skipping the exact edges 20, 21, 70, and 71.
& Assuming numeric fields will always receive clean integer inputs without testing decimals or negative signs.

Worked Walkthrough: Discount Coupon: Cart amount must be between $50.00 and $500.00 to apply coupon.
Test values: $49.99 (Min-1, error banner), $50.00 (Min, success), $50.01 (Min+1, success), $499.99 (Max-1, success), $500.00 (Max, success), $500.01 (Max+1, error banner).

3. Decision Table Testing & Requirements Traceability Matrix (RTM)

• Decision Table Testing: Used for complex business rules where multiple input conditions combine to produce specific system actions. Constructing a truth table reveals missing business logic.

• Requirements Traceability Matrix (RTM): A document linking user requirements (BRD/User Stories) to corresponding test cases and identified defects. Ensures 100% requirement coverage and helps assess impact during requirement changes.

Real-World Workplace Examples:
• **[E-commerce Free Shipping Decision Table]:** Condition 1: Order > $100? Condition 2: Prime Member? Action: Free Shipping. Rule 1: T / T -> Yes | Rule 2: T / F -> Yes | Rule 3: F / T -> Yes | Rule 4: F / F -> No ($5.99 shipping charged).

Key Standard Terminology:
• **Decision Table:** A tabular representation of logic that maps combinations of inputs to corresponding outputs.
• **RTM:** A grid that maps requirements to test cases to verify complete testing coverage from start to finish.

Common Beginner Mistakes & Interview Traps:
& Failing to update the RTM when product requirements change mid-sprint.
& Creating test cases without tracing them back to any business requirement.

3. Guided Practice Exercise

Guided Exercise: Crafting Structured Test Cases with BVA

Step-by-step conversion of an e-commerce registration form specification into formal test cases.

Feature Specification Under Test:
SPECIFICATION: Customer Registration Form
1. Full Name: 3 to 50 characters, letters only.
2. Email: Valid format (name@domain.com), must be unique in database.
3. Password: 8 to 20 characters, minimum 1 uppercase letter, 1 number, and 1 symbol.
4. Age: Must be between 18 and 65 years old inclusive.
5. Terms & Conditions checkbox: Mandatory check to submit.

Recommended Execution Steps:
• Step 1: Identify Equivalence Partitions for each field.
• Step 2: Calculate BVA test data values for Name length (2, 3, 4, 49, 50, 51) and Age (17, 18, 19, 64, 65, 66).
• Step 3: Document formal test cases using standard template fields: Test ID, Preconditions, Test Data, Steps, Expected Result.
• Step 4: Tag each test case as Positive (Happy path) or Negative (Validation failure).

4. Practical Workplace Assignment & Evaluation Rubric

Day 2 Assignment: Structured Test Case Worksheet for Registration & Checkout

Write and submit a structured test-case worksheet containing at least 8 detailed test cases covering User Registration and Discount Validation. Must include boundary value data points and negative tests.

System Under Test Brief:
Target System: Retail Store Registration & Coupon Validation. Rules: Age 18-65 inclusive, Password 8-20 chars, Discount code SAVE20 requires minimum $50.00 order subtotal.

Submission Instructions:
• Write at least 8 formal test cases.
• Include standard fields: Test Case ID, Title, Preconditions, Test Data, Step-by-Step Procedure, Expected Result, Type (Positive/Negative).
• Explicitly include Boundary Value Analysis test cases for Age (testing 17, 18, 65, 66) and Password length (testing 7 and 8 chars).
• Include negative validation test cases for uncheck terms checkbox and empty fields.

Deterministic Evaluation Rubric:
' **[Weight 30%] Minimum 8 Structured Test Cases [Requires at least 8]:** Learner documents at least 8 distinct, well-structured test cases.
' **[Weight 35%] Boundary Value Analysis Inclusion:** Includes explicit boundary testing values for numerical/length boundaries.
' **[Weight 35%] Standard Test Case Structure:** All test cases contain clear steps, preconditions, test data, and deterministic expected outcomes.

5. Industry Interview Questions & Model Answers

Q1 [Theoretical]: What is Boundary Value Analysis (BVA), and why is it preferred over random ad-hoc testing?

Model Answer:
Boundary Value Analysis is a black-box test design technique that tests values at the boundaries of equivalence partitions (Min, Min-1, Min+1, Max, Max-1, Max+1). It is preferred because empirical defect data proves developers make most mistakes at boundary conditions—such as off-by-one comparison errors (e.g. using > instead of >=)—rather than in the middle of valid ranges.

Avoid this mistake: Only testing the boundary numbers themselves without testing adjacent values (Min-1 and Max+1).

Q2 [Theoretical]: What is the purpose of a Requirements Traceability Matrix (RTM)?

Model Answer:
An RTM is a document that maps and traces user requirements with corresponding test cases and logged defects. It ensures 100% test coverage, guarantees no requirements are missed during testing, and enables accurate impact analysis when requirements change.

Interviewer Follow-Up: What are Forward Traceability and Backward Traceability?
Avoid this mistake: Thinking RTM is only used to store test execution pass/fail logs.

6. Key Takeaways & Fast Revision Notes

Core Takeaways:

- Test Scenarios define WHAT to test; Test Cases define HOW to test step-by-step.
- Equivalence Partitioning reduces redundant tests by grouping inputs into valid/invalid partitions.
- Boundary Value Analysis targets the fragile edges of input domains where bugs cluster.
- RTM links requirements to test cases to guarantee full test coverage.

Cheat Sheet / Revision Notes:

- Standard Test Case Fields: ID, Module, Title, Priority, Preconditions, Test Data, Steps, Expected Result, Actual Result, Status.
- 2-point BVA: Min, Max. 3-point BVA: Min-1, Min, Min+1, Max-1, Max, Max+1.
- Decision Table structure: Conditions (inputs) + Actions (outputs) = Business Rules.